

Calculating Normal Probabilities

Graduate Project for MATH 5021

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Background and Motivation

In his 1995 paper, Bagby discusses how to compute standard normal probabilities *without using tables*.

For a standard normal random variable X and $a > 0$, define

$$P(a) = \mathbb{P}(0 < X < a) = \frac{1}{\sqrt{2\pi}} \int_0^a e^{-x^2/2} dx.$$

This integral cannot be expressed in terms of elementary functions. At the time the paper was written, personal computers were not yet universal, so finding a **simple and accurate approximation** for $P(a)$ was very important.

From One Dimension to Two Dimensions

The **first step** in the paper is to turn a one-dimensional integral into a two-dimensional integral.

Start with

$$P(a) = \frac{1}{\sqrt{2\pi}} \int_0^a e^{-x^2/2} dx.$$

Then square both sides:

$$P(a)^2 = \left(\frac{1}{\sqrt{2\pi}} \int_0^a e^{-x^2/2} dx \right)^2 = \frac{1}{2\pi} \left(\int_0^a e^{-x^2/2} dx \right) \left(\int_0^a e^{-y^2/2} dy \right).$$

Why Does the Product Become a Double Integral?

In general,

$$\left(\int_0^a f(x) dx \right) \left(\int_0^a f(y) dy \right) = \int_0^a \int_0^a f(x)f(y) dy dx.$$

In our case, $f(t) = e^{-t^2/2}$, so

$$P(a)^2 = \frac{1}{2\pi} \int_0^a \int_0^a e^{-(x^2+y^2)/2} dy dx.$$

This is exactly the step Bagby uses to write

$$P(a)^2 = \frac{1}{2\pi} \iint_{[0,a] \times [0,a]} e^{-(x^2+y^2)/2} dx dy,$$

which prepares for a polar coordinate change and later for a new quadrature rule.

Why “Sum of Sum” Becomes a Double Sum

Consider two sums: Let

$$A = \sum_{i=1}^n a_i, \quad B = \sum_{j=1}^m b_j.$$

Then

$$A \cdot B = \left(\sum_{i=1}^n a_i \right) \left(\sum_{j=1}^m b_j \right) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j.$$

A simple example: take $n = m = 2$,

$$\begin{aligned} \left(\sum_{i=1}^2 a_i \right) \left(\sum_{j=1}^2 b_j \right) &= (a_1 + a_2)(b_1 + b_2) \\ &= a_1 b_1 + a_1 b_2 + a_2 b_1 + a_2 b_2 = \sum_{i=1}^2 \sum_{j=1}^2 a_i b_j. \end{aligned}$$

Changing to Polar Coordinates

From the previous steps, Bagby shows that

$$P(a)^2 = \frac{1}{\pi} \int_0^{\pi/4} \int_0^{a \sec \theta} e^{-r^2/2} r \, dr \, d\theta.$$

Here we have already switched to **polar coordinates**

$$x = r \cos \theta, \quad y = r \sin \theta,$$

so that $x^2 + y^2 = r^2$ and $dx \, dy = r \, dr \, d\theta$.

The limits $0 \leq \theta \leq \pi/4$ and $0 \leq r \leq a \sec \theta$ describe the region in the first quadrant under the line $y = x$ and inside the circle $r = a \sec \theta$.

From Double Integral to Bagby's Method

$$\begin{aligned} P(a)^2 &= \frac{1}{\pi} \int_0^{\pi/4} \int_0^{a \sec \theta} e^{-r^2/2} r \, dr \, d\theta \\ &= \frac{1}{\pi} \int_0^{\pi/4} \left(1 - e^{-\frac{1}{2} a^2 \sec^2 \theta} \right) d\theta. \end{aligned}$$

$$P(a)^2 = \frac{1}{4} - \frac{1}{\pi} \int_0^{\pi/4} e^{-\frac{1}{2} a^2 \sec^2 \theta} d\theta.$$

Bagby's method (core quadrature):

$$\begin{aligned} \int_{c-h}^{c+h} f(x) \, dx &\approx \frac{h}{15} \left(7f(c-h) + 16f(c) + 7f(c+h) \right) \\ &\quad - \frac{h^2}{15} \left(f'(c+h) - f'(c-h) \right). \end{aligned}$$

Quadrature Rule

We choose an **even** polynomial $K(x)$ such that

$$K^{(6)}(x) = 1, \quad K(\pm h) = 0, \quad K'(\pm h) = 0, \quad K^{(3)}(\pm h) = 0.$$

Then

$$\int_{-h}^h f(x) dx = \int_{-h}^h f(x) \cdot 1 dx = \int_{-h}^h f(x) K^{(6)}(x) dx.$$

Quadrature Rule

Set $u = f(x)$, $dv = K^{(6)}(x) dx$:

$$\int_{-h}^h f(x) K^{(6)}(x) dx = \left[K^{(5)}(x) f(x) \right]_{-h}^h - \int_{-h}^h K^{(5)}(x) f'(x) dx.$$

Repeat integration by parts 6 times:

$$\begin{aligned} \int_{-h}^h f(x) dx &= \left[K^{(5)} f - K^{(4)} f' + K^{(3)} f'' - K^{(2)} f^{(3)} + K' f^{(4)} - K f^{(5)} \right]_{-h}^h \\ &\quad + \int_{-h}^h K(x) f^{(6)}(x) dx. \end{aligned} \tag{*}$$

Quadrature Rule

From $K^{(6)}(x) = 1$ and K even (so $K^{(5)}$ and $K^{(3)}$ odd):

$$K^{(5)}(x) = x, \quad K^{(4)}(x) = \frac{1}{2}x^2 + D, \quad K^{(3)}(x) = \frac{1}{6}x^3 + Dx.$$

Use $K^{(3)}(h) = 0$:

$$0 = \frac{1}{6}h^3 + Dh \Rightarrow D = -\frac{1}{6}h^2.$$

So

$$K^{(3)}(x) = \frac{1}{6}(x^3 - h^2x), \quad K^{(4)}(x) = \frac{1}{2}x^2 - \frac{1}{6}h^2.$$

Integrate:

$$K^{(2)}(x) = \frac{1}{24}x^4 - \frac{1}{12}h^2x^2 + E, \quad K'(x) = \frac{1}{120}x^5 - \frac{1}{36}h^2x^3 + Ex.$$

Quadrature Rule

Use $K'(h) = 0$:

$$0 = \frac{1}{120}h^5 - \frac{1}{36}h^5 + Eh \Rightarrow E = \frac{7}{360}h^4.$$

Thus

$$K'(x) = \frac{1}{120}x^5 - \frac{1}{36}h^2x^3 + \frac{7}{360}h^4x.$$

Integrate and use $K(h) = 0$:

$$K(x) = \frac{1}{720} \left(x^6 - 5h^2x^4 + 7h^4x^2 - 3h^6 \right) = \frac{1}{720} (x^2 - h^2)^2 (x^2 - 3h^2).$$

Quadrature Rule

Recall (★):

$$\int_{-h}^h f(x) dx = \left[K^{(5)}f - K^{(4)}f' + K^{(3)}f'' - K^{(2)}f^{(3)} + K'f^{(4)} - Kf^{(5)} \right]_{-h}^h \\ + \int_{-h}^h K(x)f^{(6)}(x) dx.$$

Therefore,

$$\int_{-h}^h f(x) dx = h(f(h) + f(-h)) - \frac{h^2}{3}(f'(h) - f'(-h)) \\ + \frac{h^4}{45}(f^{(3)}(h) - f^{(3)}(-h)) + \int_{-h}^h K(x)f^{(6)}(x) dx.$$

Shift x by c :

$$\int_{-h}^h f(x+c) dx = \int_{c-h}^{c+h} f(u) du.$$

So replace $f(x)$ by $f(x+c)$ in the formula on $[-h, h]$:

$$\begin{aligned} \int_{c-h}^{c+h} f(x) dx &= h(f(c+h) + f(c-h)) - \frac{h^2}{3}(f'(c+h) - f'(c-h)) \\ &\quad + \frac{h^4}{45}(f^{(3)}(c+h) - f^{(3)}(c-h)) \\ &\quad + \frac{1}{720} \int_{-h}^h (x^2 - h^2)^2 (x^2 - 3h^2) f^{(6)}(c+x) dx. \end{aligned}$$

Quadrature Rule

Apply to the left half: $\int_{c-h}^c f(x) dx$.

Here the midpoint is $c - \frac{h}{2}$ and the half-width is $\frac{h}{2}$, so we use the one-step formula with $h \mapsto \frac{h}{2}$ and $c \mapsto c - \frac{h}{2}$:

$$\begin{aligned} \int_{c-h}^c f(x) dx &= \frac{h}{2} \left(f(c) + f(c-h) \right) - \frac{h^2}{12} \left(f'(c) - f'(c-h) \right) \\ &\quad + \frac{h^4}{720} \left(f^{(3)}(c) - f^{(3)}(c-h) \right) \\ &\quad + \frac{1}{720} \int_{-h/2}^{h/2} \left(x^2 - \frac{h^2}{4} \right)^2 \left(x^2 - \frac{3h^2}{4} \right) f^{(6)} \left(c - \frac{h}{2} + x \right) dx. \end{aligned}$$

Right half: similarly, $\int_c^{c+h} f(x) dx$ has an analogous formula (replace $c - \frac{h}{2}$ by $c + \frac{h}{2}$).

Quadrature Rule

Two-step formula: add the left half and the right half.

$$\begin{aligned}\int_{c-h}^{c+h} f(x) dx &= \frac{h}{2} \left(f(c+h) + 2f(c) + f(c-h) \right) \\ &\quad - \frac{h^2}{12} \left(f'(c+h) - f'(c-h) \right) \\ &\quad + \frac{h^4}{720} \left(f^{(3)}(c+h) - f^{(3)}(c-h) \right) \\ &\quad + (\text{remainder term}).\end{aligned}$$

Quadrature Rule:**

$$\text{Rule}^{**} = \frac{16}{15} (\text{Two-step}) - \frac{1}{15} (\text{One-step}).$$

Quadrature Rule

$$\begin{aligned}\int_{-h}^h f(c+x) dx &= \frac{16}{15} \left\{ \frac{h}{2} (f(c+h) + 2f(c) + f(c-h)) - \frac{h^2}{12} (f'(c+h) - f'(c-h)) \right. \\ &\quad + \frac{h^4}{720} (f^{(3)}(c+h) - f^{(3)}(c-h)) \\ &\quad + \frac{1}{720} \int_{-h/2}^{h/2} (f^{(6)}(c - \frac{h}{2} + x) + f^{(6)}(c + \frac{h}{2} + x)) \left(x^2 - \frac{h^2}{4}\right)^2 \left(x^2 - \frac{3h^2}{4}\right) dx \Big\} \\ &- \frac{1}{15} \left\{ h(f(c+h) + f(c-h)) - \frac{h^2}{3} (f'(c+h) - f'(c-h)) \right. \\ &\quad + \frac{h^4}{45} (f^{(3)}(c+h) - f^{(3)}(c-h)) \\ &\quad + \frac{1}{720} \int_{-h}^h f^{(6)}(c+x) (x^2 - h^2)^2 (x^2 - 3h^2) dx \Big\}.\end{aligned}$$

Application

For $n \in \mathbb{N}$, define

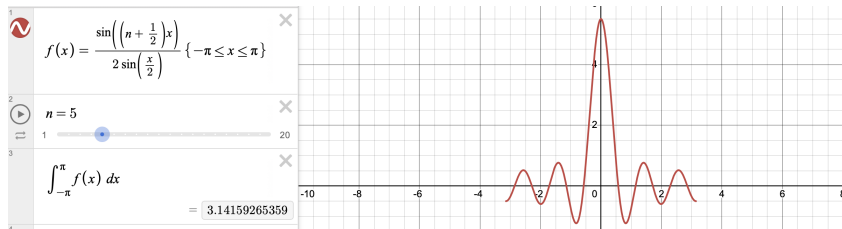
$$f_n(x) = \begin{cases} n + \frac{1}{2}, & x = 0, \\ \frac{\sin((n + \frac{1}{2})x)}{2 \sin(x/2)}, & x \neq 0. \end{cases}$$

We approximate

$$I = \int_{-\pi}^{\pi} f_n(x) dx \quad (\text{true value } I = \pi).$$

Goal: use the *composite Bagby quadrature rule* to compute $I \approx \pi$.

application



Application

Bagby's quadrature rule on $[a, b]$:

$$\int_a^b f(x) dx \approx \frac{b-a}{30} \left(7f(a) + 16f\left(\frac{a+b}{2}\right) + 7f(b) \right) - \frac{(b-a)^2}{60} \left(f'(b) - f'(a) \right).$$

partition $[-\pi, \pi]$ into M equal subintervals:

$$x_j = -\pi + j\Delta, \quad \Delta = \frac{2\pi}{M}, \quad j = 0, 1, \dots, M.$$

For each $[x_j, x_{j+1}]$ with midpoint $m_j = \frac{x_j + x_{j+1}}{2}$, set

$$I_j = \frac{\Delta}{30} \left(7f(x_j) + 16f(m_j) + 7f(x_{j+1}) \right) - \frac{\Delta^2}{60} \left(f'(x_{j+1}) - f'(x_j) \right),$$

then

$$I_M = \sum_{j=0}^{M-1} I_j \approx \int_{-\pi}^{\pi} f(x) dx.$$

Application

We need f_n and f'_n .

For $x \neq 0$, let $A = n + \frac{1}{2}$ and

$$f_n(x) = \frac{\sin(Ax)}{2 \sin(x/2)}.$$

Removable value at $x = 0$ (L'Hôpital):

$$\lim_{x \rightarrow 0} \frac{\sin(Ax)}{2 \sin(x/2)} \stackrel{0/0}{=} \lim_{x \rightarrow 0} \frac{A \cos(Ax)}{\cos(x/2)} = A, \quad \Rightarrow \quad f_n(0) = A.$$

Since f_n is even, $f'_n(0) = 0$.

Derivative for $x \neq 0$ (quotient rule):

$$f'_n(x) = \frac{2A \cos(Ax) \sin(x/2) - \sin(Ax) \cos(x/2)}{4 \sin^2(x/2)}.$$

Application

Bagby with $M = 2$ (two panels).

Let $A = n + \frac{1}{2}$ and

$$f_n(x) = \begin{cases} A, & x = 0, \\ \frac{\sin(Ax)}{2 \sin(x/2)}, & x \neq 0. \end{cases}$$

Partition $[-\pi, \pi]$ into $M = 2$ equal subintervals:

$$\Delta = \frac{2\pi}{2} = \pi, \quad x_0 = -\pi, \quad x_1 = 0, \quad x_2 = \pi.$$

Midpoints:

$$m_0 = \frac{x_0 + x_1}{2} = -\frac{\pi}{2}, \quad m_1 = \frac{x_1 + x_2}{2} = \frac{\pi}{2}.$$

Bagby rule on $[a, b]$:

$$\int_a^b f(x) dx \approx \frac{b-a}{30} \left(7f(a) + 16f\left(\frac{a+b}{2}\right) + 7f(b) \right) - \frac{(b-a)^2}{60} \left(f'(b) - f'(a) \right).$$

Application

Compute I_0 on $[-\pi, 0]$ and I_1 on $[0, \pi]$.

Since f_n is even, $f'_n(0) = 0$, and also $f'_n(\pm\pi) = 0$. Hence the derivative correction vanishes on both panels.

For $\Delta = \pi$:

$$I_0 = \frac{\pi}{30} \left(7f_n(-\pi) + 16f_n(-\frac{\pi}{2}) + 7f_n(0) \right), \quad I_1 = \frac{\pi}{30} \left(7f_n(0) + 16f_n(\frac{\pi}{2}) + 7f_n(\pi) \right).$$

Values needed:

$$f_n(0) = A, \quad f_n(\pi) = f_n(-\pi) = \frac{\sin(A\pi)}{2} = \frac{(-1)^n}{2},$$

$$f_n\left(\frac{\pi}{2}\right) = f_n\left(-\frac{\pi}{2}\right) = \frac{\sin(A\pi/2)}{2\sin(\pi/4)} = \frac{\sin(A\pi/2)}{\sqrt{2}}.$$

Combine the two panels (closed form for $M = 2$).

Using $f_n(-\pi) = f_n(\pi)$ and $f_n(-\frac{\pi}{2}) = f_n(\frac{\pi}{2})$,

$$\begin{aligned} I_2 &:= I_0 + I_1 = \frac{\pi}{30} \left(14f_n(\pi) + 32f_n\left(\frac{\pi}{2}\right) + 14f_n(0) \right) \\ &= \frac{\pi}{30} \left(14A + 7(-1)^n + \frac{32}{\sqrt{2}} \sin\left(\frac{A\pi}{2}\right) \right). \end{aligned}$$

Example ($n = 20$, so $A = 20.5$):

$$\sin\left(\frac{A\pi}{2}\right) = \sin(10.25\pi) = \sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2},$$

$$I_2 = \frac{\pi}{30}(287 + 7 + 16) = \frac{31}{3}\pi \approx 32.4697.$$

Quadrature Rule

Numerical results (example: $n = 20$).

True value: $\pi \approx 3.141592653590$.

M	Composite Bagby I_M	$ I_M - \pi $
10	9.005898940291	5.864306286701
20	2.722713633111	0.418879020479
40	3.141592653590	≈ 0
80	3.141592653590	≈ 0

Conclusion: as M increases (smaller step size), the composite Bagby rule converges rapidly to π for this integral.